Instrumental Variables (IV) Estimation

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Motivation and Learning Objectives

Instrumental Variables (IV) estimation is a powerful technique for estimating causal effects when randomization is not possible due to confounding. This method allows us to identify the causal effect of a treatment on an outcome by using an instrument that affects treatment but is not directly related to the outcome. In this lesson, we will:

- Understand the rationale behind IV estimation and its assumptions.
- Derive the IV estimand using the potential outcomes framework.
- Explain the relationship between observed outcomes and potential outcomes.
- Explore the mathematical formulation of IV estimation.
- Explain the expected value of observed outcomes given an instrument.
- Understand the core assumption of consistency in causal inference.
- Apply these concepts to practical examples.

Instrumental Variables (IV) Estimation

The IV estimand calculates the Local Average Treatment Effect (LATE)—the causal effect of a treatment for the specific subgroup of individuals whose treatment status is changed by the instrument.

1. Rationale and Potential Outcomes Framework

The potential outcomes framework helps us define causal effects precisely. For each individual i, we define:

- Y_i^1: The potential outcome if the individual receives the treatment ($A = 1$).
- Y_i^0: The potential outcome if the individual does not receive the treatment ($A = 0$).
- The individual-level causal effect is $Y_i^1 - Y_i^0$. We can only ever observe one of these for any given person.

Similarly, we define potential treatment assignments based on the instrument, Z:

- A_i^1: The treatment the individual would take if the instrument is present ($Z = 1$).
- A_i^0: The treatment the individual would take if the instrument is absent ($Z = 0$).

The challenge of confounding means that the groups who choose $A = 1$ and $A = 0$ are different in ways that also affect the outcome. The instrument (Z) helps us overcome this by creating an "as-if" random reason for changing treatment status.

IV Assumptions in Potential Outcomes

1. **Relevance**: $\mathbb{E}[A^1] - \mathbb{E}[A^0] \neq 0$. The instrument actually affects treatment uptake on average.
2. **Independence**: $Z \perp\!\!\!\perp (Y^1, Y^0, A^1, A^0)$. The instrument is as-if randomly assigned and is therefore independent of an individual's potential outcomes and potential treatment choices. This is the key assumption that replaces the "no unmeasured confounders" idea.
3. **Exclusion Restriction**: $Y_i^{a,z} = Y_i^a$ for $a \in \{0, 1\}$. An individual's potential outcome depends only on the treatment they receive (a), not on the value of the instrument (z). The instrument works *only* through the treatment.
4. **Monotonicity**: $A_i^1 \geq A_i^0$ for all individuals. The instrument encourages (or has no effect on) treatment uptake for everyone; it never discourages it. This assumption rules out the existence of "defiers."

2. Step-by-Step Derivation

The derivation elegantly shows how the IV formula isolates the average treatment effect for a specific group.

Step 1: Decomposing the Population

Under the monotonicity assumption, we can classify everyone into three groups based on their potential treatment choices. This is called principal stratification.

- Compliers (c): Individuals who take the treatment if encouraged but not otherwise. ($A_i^0 = 0, A_i^1 = 1$).
- Always-Takers (a): Individuals who take the treatment regardless of the instrument. ($A_i^0 = 1, A_i^1 = 1$).
- Never-Takers (n): Individuals who do not take the treatment regardless of the instrument. ($A_i^0 = 0, A_i^1 = 0$).

Let π_c, π_a, π_n be the proportion of each group in the population.

Step 2: Analyzing the Numerator: $\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]$

First, let's expand the term $\mathbb{E}[Y|Z=1]$. The observed outcome Y when $Z = 1$ is determined by the treatment individuals take when $Z = 1$, which is A^1. So, $Y = Y(A^1)$.

$$\mathbb{E}[Y|Z=1] = \mathbb{E}\left[Y(A^1)|Z=1\right]$$

By the independence assumption, we can drop the conditioning on $Z = 1$:

$$\mathbb{E}[Y|Z=1] = \mathbb{E}\left[Y(A^1)\right]$$

Using the law of total expectation, we can write this as a weighted average across our three groups:

$$\mathbb{E}\left[Y(A^1)\right] = \mathbb{E}\left[Y(A^1)|\text{type} = c\right]\pi_c + \mathbb{E}\left[Y(A^1)|\text{type} = n\right]\pi_n + \mathbb{E}\left[Y(A^1)|\text{type} = a\right]\pi_a$$

Now, substitute the known treatment status for each group when $Z = 1$:

- For Compliers, $A^1 = 1$, so the term becomes $\mathbb{E}\left[Y^1|\text{type} = c\right]\pi_c$.
- For Never-Takers, $A^1 = 0$, so the term becomes $\mathbb{E}\left[Y^0|\text{type} = n\right]\pi_n$.
- For Always-Takers, $A^1 = 1$, so the term becomes $\mathbb{E}\left[Y^1|\text{type} = a\right]\pi_a$.

So, $\mathbb{E}[Y|Z=1] = \mathbb{E}\left[Y^1|c\right]\pi_c + \mathbb{E}\left[Y^0|n\right]\pi_n + \mathbb{E}\left[Y^1|a\right]\pi_a$.

Now we do the same for $\mathbb{E}[Y|Z=0] = \mathbb{E}\left[Y(A^0)\right]$:

$$\mathbb{E}\left[Y(A^0)\right] = \mathbb{E}\left[Y(A^0)|c\right]\pi_c + \mathbb{E}\left[Y(A^0)|n\right]\pi_n + \mathbb{E}\left[Y(A^0)|a\right]\pi_a$$

- For Compliers, $A^0 = 0$, so the term becomes $\mathbb{E}\left[Y^0|\text{type} = c\right]\pi_c$.
- For Never-Takers, $A^0 = 0$, so the term becomes $\mathbb{E}\left[Y^0|\text{type} = n\right]\pi_n$.
- For Always-Takers, $A^0 = 1$, so the term becomes $\mathbb{E}\left[Y^1|\text{type} = a\right]\pi_a$.

So, $\mathbb{E}[Y|Z=0] = \mathbb{E}\left[Y^0|c\right]\pi_c + \mathbb{E}\left[Y^0|n\right]\pi_n + \mathbb{E}\left[Y^1|a\right]\pi_a$.

Finally, we subtract the two expressions. Notice that the terms for always-takers and never-takers are identical in both equations, so they cancel out perfectly!

$$\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0] = (\mathbb{E}\left[Y^1|c\right]\pi_c) - (\mathbb{E}\left[Y^0|c\right]\pi_c)$$

$$\implies \mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0] = (\mathbb{E}\left[Y^1|c\right] - \mathbb{E}\left[Y^0|c\right]) \cdot \pi_c$$

The numerator is the average treatment effect for compliers multiplied by the proportion of compliers.

Step 3: Analyzing the Denominator: $\mathbb{E}[A|Z=1] - \mathbb{E}[A|Z=0]$

This step is more straightforward. Using the same logic:

$$\mathbb{E}[A|Z=1] - \mathbb{E}[A|Z=0] = \mathbb{E}\left[A^1\right] - \mathbb{E}\left[A^0\right]$$

Let's expand each term across our three groups:

$$\mathbb{E}\left[A^1\right] = \mathbb{E}\left[A^1|c\right]\pi_c + \mathbb{E}\left[A^1|n\right]\pi_n + \mathbb{E}\left[A^1|a\right]\pi_a$$

$$\mathbb{E}\left[A^1\right] = (1)\pi_c + (0)\pi_n + (1)\pi_a = \pi_c + \pi_a$$

And:

$$\mathbb{E}\left[A^0\right] = \mathbb{E}\left[A^0|c\right]\pi_c + \mathbb{E}\left[A^0|n\right]\pi_n + \mathbb{E}\left[A^0|a\right]\pi_a$$

$$\mathbb{E}\left[A^0\right] = (0)\pi_c + (0)\pi_n + (1)\pi_a = \pi_a$$

Subtracting the two gives:

$$\mathbb{E}[A|Z=1] - \mathbb{E}[A|Z=0] = (\pi_c + \pi_a) - (\pi_a) = \pi_c$$

The denominator is simply the proportion of compliers in the population.

Step 4: Combining the Numerator and Denominator

Now we assemble the final IV estimand ratio:

$$\text{IV Estimand} = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[A|Z=1] - \mathbb{E}[A|Z=0]} = \frac{\left(\mathbb{E}\left[Y^1|c\right] - \mathbb{E}\left[Y^0|c\right]\right)\cdot\pi_c}{\pi_c}$$

The π_c terms cancel, leaving our final result:

$$\text{IV Estimand} = \mathbb{E}\big[Y^1|c\big] - \mathbb{E}\big[Y^0|c\big]$$

This proves that under the IV assumptions, the formula precisely calculates the Local Average Treatment Effect (LATE): the average causal effect of the treatment *only* for the subpopulation of compliers.

The relationship between what we can see (observed outcomes) and what could have happened (potential outcomes) is the foundation of modern causal inference. It allows us to use real-world data to answer "what if" questions.

The Core Assumption: Consistency

The primary rule connecting observed and potential outcomes is the consistency assumption. It states that an individual's observed outcome is simply their potential outcome corresponding to the treatment they actually received.

- For the Outcome (Y): If an individual receives treatment $A = a$, their observed outcome Y is equal to their potential outcome $Y(a)$.
  - If you take the medicine $(A = 1)$, the headache level you experience (Y) is precisely your potential headache level had you taken the medicine (Y^1).
  - If you don't take the medicine $(A = 0)$, the headache level you experience (Y) is your potential headache level had you not taken it (Y^0).
- For the Treatment (A): The same logic applies if there's an instrument (Z) influencing the treatment. If an individual is exposed to instrument level $Z = z$, their observed treatment status A is equal to their potential treatment status $A(z)$.
  - If you are encouraged by your doctor to get a vaccine $(Z = 1)$, your observed vaccination status (A) is your potential status had you been encouraged (A^1).

Without this assumption, we couldn't link the data we collect to the causal effects we want to estimate. Consistency is one half of the broader Stable Unit Treatment Value Assumption (SUTVA), with the other half being "no interference" between individuals.

Mathematical Formulation

We can express these relationships with simple formulas. For a binary treatment $(A \in \{0, 1\})$:

Outcome Y

The observed outcome Y is a "switched" version of the two potential outcomes, where the switch is controlled by the observed treatment A:

$$Y = A \cdot Y^1 + (1 - A) \cdot Y^0$$

How it works:

- If an individual gets the treatment, $A = 1$. The formula becomes $Y = (1) \cdot Y^1 + (1 - 1) \cdot Y^0 = Y^1$.
- If an individual does not get the treatment, $A = 0$. The formula becomes $Y = (0) \cdot Y^1 + (1 - 0) \cdot Y^0 = Y^0$.

Treatment A (with an Instrument Z)

Similarly, the observed treatment A is a switched version of the potential treatment statuses, controlled by the observed instrument Z:

$$A = Z \cdot A^1 + (1 - Z) \cdot A^0$$

How it works:

- If an individual is exposed to the instrument, $Z = 1$. The formula becomes $A = (1) \cdot A^1 + (1 - 1) \cdot A^0 = A^1$.
- If an individual is not exposed to the instrument, $Z = 0$. The formula becomes $A = (0) \cdot A^1 + (1 - 0) \cdot A^0 = A^0$.

These equations are the fundamental link that allows us to take observable quantities (Y, A, Z) and, under further assumptions like ignorability, make statements about the unobservable causal effects derived from potential outcomes $(Y^1 - Y^0)$.

Explaining Expected Value of Observed Y given Observed Instrument

This equality is established by applying the consistency assumption twice, which is the fundamental rule linking observed data to the potential outcomes framework.

Here is a detailed step-by-step explanation.

Step 1: The Starting Point

We begin with the left side of the equation:

$$\mathbb{E}[Y|Z = 1]$$

This expression represents the expected (or average) observed outcome Y among the subgroup of individuals who received the instrument at level 1 ($Z = 1$).

Step 2: Applying Consistency to the Outcome (Y)

The first rule we use is the consistency assumption for the outcome. This rule states that an individual's observed outcome (Y) is their potential outcome corresponding to the treatment level (A) they actually received.

- Rule: $Y = Y^a$
- Application: We can substitute Y^a for Y inside our expectation, as they are defined to be equivalent.

$$\mathbb{E}[Y|Z = 1] = \mathbb{E}[Y^a|Z = 1]$$

This expression now reads: "The expected potential outcome, evaluated at the observed treatment A, for the group where $Z = 1$."

Step 3: Applying Consistency to the Treatment (A)

Next, we focus on the observed treatment, A, specifically for the group conditioned on $Z = 1$. The consistency assumption for the treatment states that an individual's observed treatment status (A) is their potential treatment status corresponding to the instrument level (Z) they were actually exposed to.

- Rule: $A = A(Z)$
- Application: For the specific subgroup where $Z = 1$, this rule simplifies to: $A = A^1$. This means that for everyone in this group, their observed treatment is equal to their potential treatment had they received the instrument.

Step 4: Final Substitution

Now we can substitute the result from Step 3 into the expression from Step 2. Since we are working entirely within the subgroup where A is equivalent to A^1, we can replace A with A^1 inside the potential outcome function.

$$\mathbb{E}[Y^a|Z = 1] = \mathbb{E}\big[Y(A^1)|Z = 1\big]$$

By chaining these steps together, we show the full equality:

$$\mathbb{E}[Y|Z = 1] \xrightarrow{\text{Consistency on Y}} \mathbb{E}[Y^a|Z = 1] \xrightarrow{\text{within the Z=1 group}}_{\text{Consistency on A}} \mathbb{E}\big[Y(A^1)|Z = 1\big]$$

This final expression, $\mathbb{E}\big[Y(A^1)|Z = 1\big]$, represents the expected potential outcome that results from the potential treatment an individual would choose if exposed to the instrument.